Individual Contributions Report: Exploratory Data Analysis Lab

Course: ITAI 1371 – Principles of Machine Learning

Project: Module 04 Lab – Exploratory Data Analysis & Pipeline Integration

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Lead Coordinator: Arinze Naphitaly Okafor

1. Project Leadership & Team Management

As the project lead and coordinator for this lab, I took ownership of our team's workflow and final deliverables. My main responsibility outside of coding was keeping the team on schedule, tracking individual progress, and making sure everyone pulled their weight.

To keep us on track, I:

- Established direct communication channels and assigned specific tasks based on our lab requirements.
- Followed up regularly with each team member to check on their progress, answer questions, and ensure workload was split fairly.
- Compiled and formatted all code cells, visual plots, analysis sections, and research notes into a clean, unified submission file.

2. Technical Code & Written Contributions

My Work (Arinze Naphitaly Okafor - Lead & Coordinator)

- **Code Implementation:** I wrote and executed the code for both experimentation sections in our notebook.
    - **Experiment 1 (Port of Embarkation):** I built a sns.countplot() using hue='Survived' to analyze how survival rates varied across Cherbourg (C), Queenstown (Q), and Southampton (S).

  - **Experiment 2 (Fare vs. Survival):** I created log-scaled boxplots (sns.boxplot() with plt.yscale('log')) to clean up the extreme fare outliers and analyze the fare distribution between survivors and non-survivors.

- **Written Analysis & Knowledge Check:** I authored the analytical insights for the experiments, explaining how higher survival rates at Cherbourg correlated with a larger share of 1st-class ticket holders. I also completed the Knowledge Check section, breaking down the core goals of EDA, survival priority factors (1st-class females and young children), and why visualizations reveal critical patterns that raw summary stats miss.
- **Research Report:** I researched, structured, and wrote our team's EDA summary report, detailing our data cleaning strategies (imputing missing ages, handling cabin column sparsity) and showing how EDA trends inform model selection and evaluation metrics down the line.

Hamza Ajmal

Contribution Journal: Module 04 Lab, Exploratory Data Analysis

ITAI 1371 - Introduction to Machine Learning

Summary of my contribution

I completed this lab and its deliverables myself. My work covered running and understanding the notebook, diagnosing and fixing the errors that prevented it from running, completing both student experiments, answering the Knowledge Check, and preparing the submission files. I am describing each part below in my own words.

Running and understanding the notebook

I worked through the notebook in order: loading the Titanic dataset, reading the descriptive statistics, and studying the four guided visualizations (survival counts, survival by class, survival by gender, and the age histograms). For each one I made sure I understood what the plot was meant to show and how it connected to the overall question of what predicts survival.

Finding and fixing the notebook errors

When I ran the notebook, most cells failed. The first cell raised a SyntaxError and the rest then failed with NameErrors because the imports never ran. I traced the cause of lost line breaks in the code cells, which turned each cell into a single line of code, and to an escaped newline inside a print statement that had been split across two lines. I restored the line breaks without changing any of the lab's text, structure, or my own experiment cells, then re-ran the notebook so all outputs are visible.

Experiments

Experiment 1 (Port of Embarkation): I wrote a count plot of survival by port, then added survival rates per port and a crosstab of port against passenger class. This let me see that Cherbourg had the highest survival rate but also the most 1st-class passengers, so I concluded that port is likely a proxy for class rather than a direct cause.

Experiment 2 (Fare vs. Survival): I created boxplots of fare by survival status on both a normal and a log scale and compared median fares. This showed that survivors generally paid higher fares and that the fare distribution is heavily skewed.

Knowledge Check, journals, and submission

I answered the three Knowledge Check questions in the notebook's markdown cell, wrote this contribution journal and the reflective journal, exported the executed notebook to PDF, and checked that the plots and code were legible. I also prepared the files for the GitHub repository submission as the course instructions require.

Johna Matthews

Contribution Journal

Introduction

For this project, my contribution focused on working through and understanding the Exploratory Data Analysis (EDA) of the Titanic dataset. My role was to examine the data, run the provided analysis, interpret the results, and identify the main patterns shown by the statistics and visualizations. I also focused on understanding why each step of EDA is important before developing a machine learning model.

My Specific Contributions

I worked with the Titanic dataset by loading, inspecting, and analyzing the data. I used descriptive statistics to understand numerical information and examined survival patterns based on passenger class, gender, and age. I found that more passengers died than survived, while females and passengers in higher classes generally had better survival rates. I also interpreted the graphs and statistics and explained the main findings in my own words.

My Learning and Reflection

Through my contribution, I developed a better understanding of how EDA is used to investigate a dataset before modeling. I learned that numerical summaries are useful for understanding the general characteristics of data, but visualizations make relationships and patterns much easier to recognize. For example, the passenger-class and gender visualizations made the differences in survival immediately clear. I also learned that missing values must be identified early because data quality can affect later analysis.

Another important part of my contribution was interpreting the results rather than simply producing graphs. The analysis showed associations between survival and variables such as passenger class, gender, and age. I understand that these relationships should be interpreted carefully and should not automatically be treated as proof that one variable caused another.

Limitations and Future Contribution

In future project work, I would strengthen my contribution by creating additional visualizations independently, testing relationships that are not already provided, and documenting the findings clearly for the team.

Conclusion

Overall, my contribution to the project was centered on exploring the Titanic dataset, understanding the EDA process, interpreting descriptive statistics and visualizations, and identifying meaningful patterns in passenger survival. This work helped me improve my Python-based data-analysis skills and my ability to communicate what data findings mean.

Jonathon Wilturner

Contribution Journal: Module 04 Lab, Exploratory Data Analysis

ITAI 1371 - Introduction to Machine Learning

My contribution summary

I completed this lab and all of its deliverables on my own. My work included running and understanding the notebook, finding and fixing the errors that were blocking it from running, completing both student experiments, answering the Knowledge Check questions, and putting together the submission files. I'm breaking each part down below in my own words.

My understanding of running the notebook

I went through the notebook in order, loading the Titanic dataset, reading through the descriptive statistics, and working through the four guided visualizations: survival counts, survival by class, survival by gender, and the age histograms. For each one, I made sure I actually understood what the plot was showing and how it tied back to the bigger question, what predicts survival.

Fixing notebook errors and how I found them

When I first ran the notebook, almost every cell failed. The first cell threw a SyntaxError, and everything after it failed with NameErrors because the import cell never actually ran. I traced it back to two things: the line breaks inside the code cells had gotten lost, which turned whole cells into one long line of code, and an escaped newline inside a print statement that had been split across two lines. I fixed the line breaks without touching any of the lab's original text, structure, or my own experiment cells, then reran the whole notebook so every output shows up correctly.

Experiments

Experiment 1 (Port of Embarkation): I built a count plot of survival by port, then added survival rates by port and a crosstab comparing port against passenger class. That showed Cherbourg had

the highest survival rate, but also the most 1st-class passengers, so I concluded port was likely standing in for class rather than causing survival on its own.

Experiment 2 (Fare vs. Survival): I made boxplots of fare by survival status, one on a normal scale and one on a log scale, and compared median fares between the two groups. This showed survivors generally paid higher fares, and that the fare distribution is heavily skewed.

Knowledge Check, journals, and submission

I answered the three Knowledge Check questions directly in the notebook's markdown cell, wrote this contribution journal along with the reflective journal, exported the fully run notebook to PDF, and checked that all the plots and code were clear and legible. I also set up the files for the GitHub repo submission the way the course instructions require.